Assignment 1

EE5141 - Wireless Communication

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1 Introduction

This report contains the answers to assignment 1. It only contains my plots and approaches to the problems. I have attached my matlab code which contains 2 files with this result pdf.

2 Time selective fading with no ISI

2.1 Levinson Algorithm

For a filter of order p, the Levinson-Durbin algorithm efficiently solves the following equations. This relates the autocorrelation matrix, the predictor coefficients, and the autocorrelation vector. For a theoretical Rayleigh fading channel, the discrete autocorrelation sequence $R[k]$ is defined by the zeroth-order Bessel function of the first kind:

$$R[k] = J_0(2\pi f_d k T_s)$$

where f_d is the maximum Doppler frequency and T_s is the sampling period. The matrix formulation expands as follows:

$$\begin{bmatrix} R[0] & R[1] & \cdots & R[p-1] \\ R[1] & R[0] & \cdots & R[p-2] \\ \vdots & \vdots & \ddots & \vdots \\ R[p-1] & R[p-2] & \cdots & R[0] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = - \begin{bmatrix} R[1] \\ R[2] \\ \vdots \\ R[p] \end{bmatrix}$$

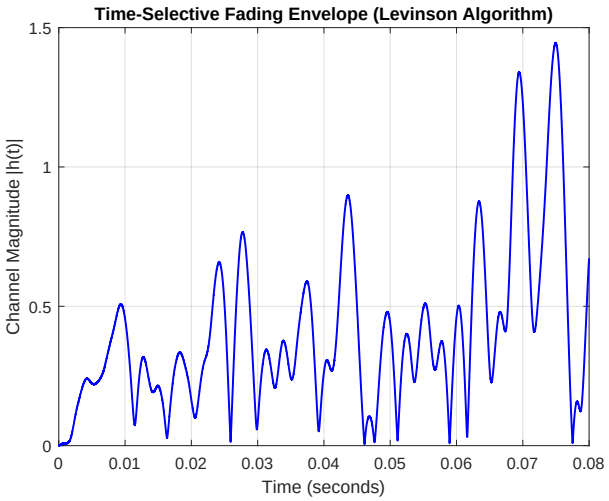
These solved coefficients a_k form the denominator polynomial $A(z) = 1 + \sum_{k=1}^{p} a_k z^{-k}$ of an all-pole Autoregressive (AR) filter.

By passing white noise $w[n]$ through this filter, we generate the channel realization $h[n]$. Conceptually, this relationship is defined by the transfer function:

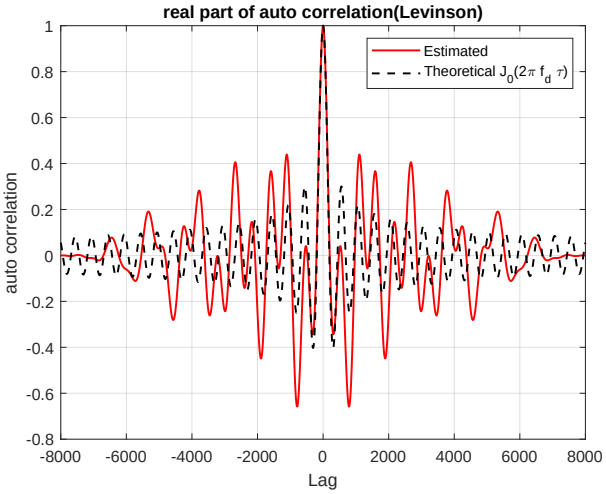
$$H(z) = \frac{W(z)}{A(z)} \implies h[n] \leftarrow \frac{w[n]}{A(z)}$$

In the time domain, this expands to the standard difference equation:

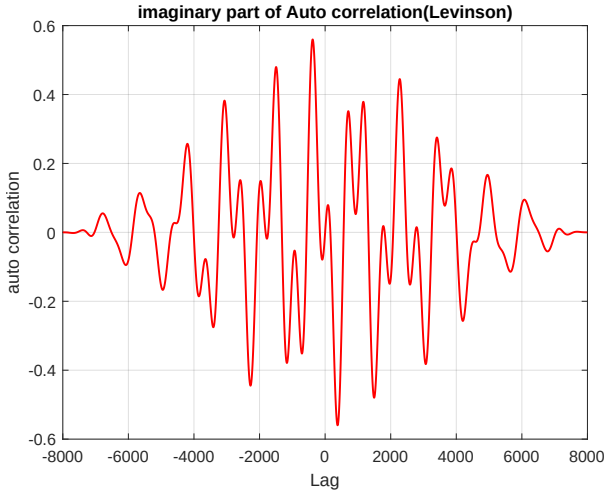
$$h[n] = -\sum_{k=1}^{p} a_k h[n-k] + w[n]$$



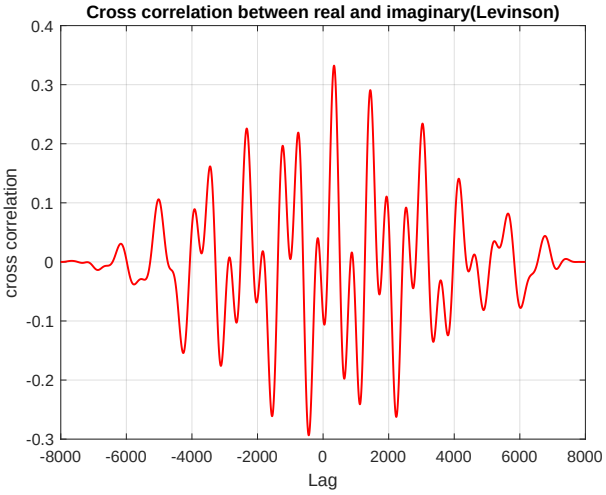
(a) Fading channel magnitude (v = 30m/s)



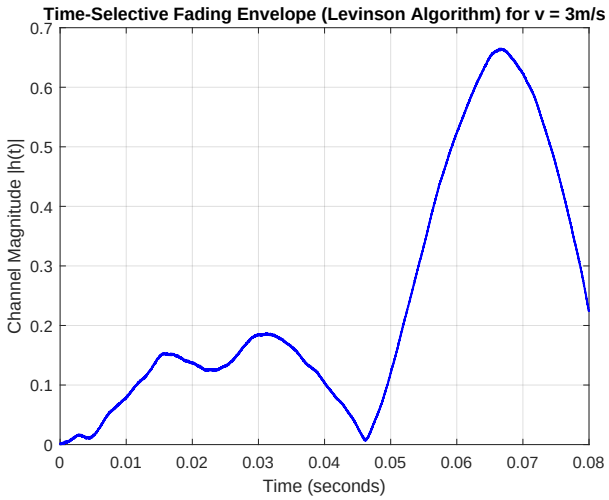
(b) Autocorrelation (Real part)



(c) Autocorrelation (Imaginary part)



(d) Cross Correlation



(e) Fading channel magnitude (v = 3m/s)

Figure 1: Analysis of time-varying fading channel magnitude and correlation properties across different mobilities using Levinson's algorithm.

2.2 Smiths Model

Smith's model is based on the fundamental linear system relationship where the output Power Spectral Density (PSD) is determined by the input PSD and the filter's frequency response:

$$S_{out}(f) = S_{in}(f)|H(f)|^2$$

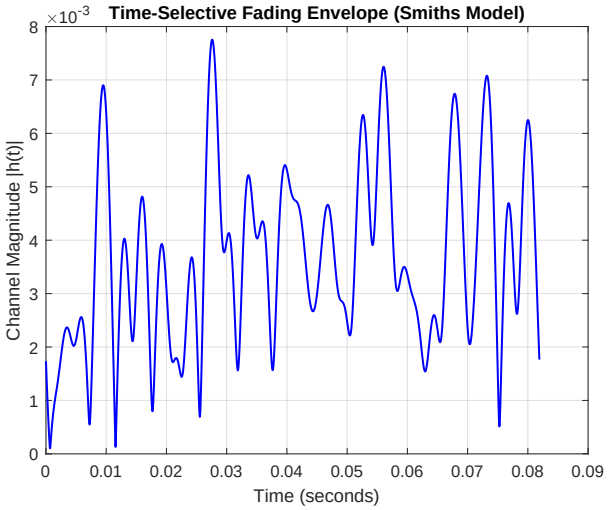
We define the input as a complex Gaussian random process, $n(t) = n_I(t) + jn_Q(t)$. Both the in-phase ($n_I(t)$) and quadrature ($n_Q(t)$) components are generated with a variance of 1. Consequently, the net variance (total power) of this time-domain input signal is 2.

Taking the discrete Fourier transform of these independent branches yields conjugate symmetric Gaussian processes in the frequency domain, scaled to a variance of N. To shape this flat spectrum into the desired fading profile, we multiply the frequency-domain noise by the filter response:

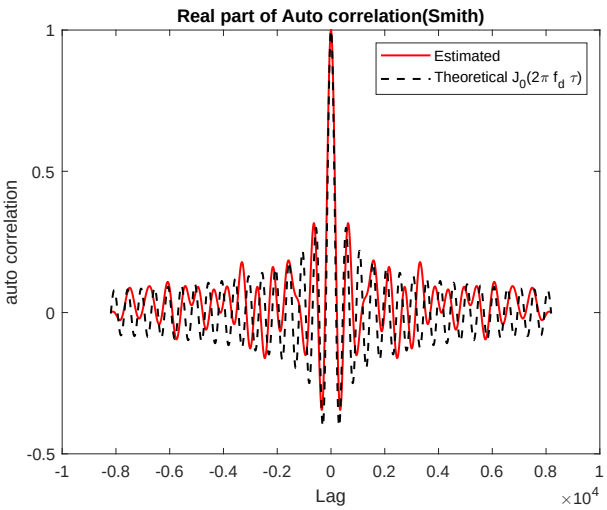
$$H(f) = \sqrt{\frac{S_{Jakes}(f)}{2}}$$

The factor of 2 in the denominator compensates for the input noise variance of 2. When we evaluate the magnitude squared of this filter, $|H(f)|^2$, the factor of 2 perfectly cancels out the input power, ensuring that the final output PSD exactly matches the theoretical Jakes PSD ($S_{Jakes}(f)$).

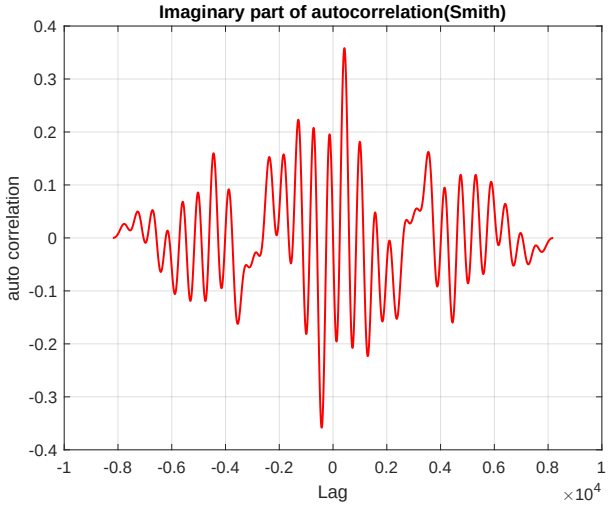
NOTE: For the given fs and N=8192 the number of samples for fd corresponding to 3m/s was only 1 which was way to less for the simulation. So I reduced fs to fs/10 and truncated the output time domain to get a corresponding comparison.



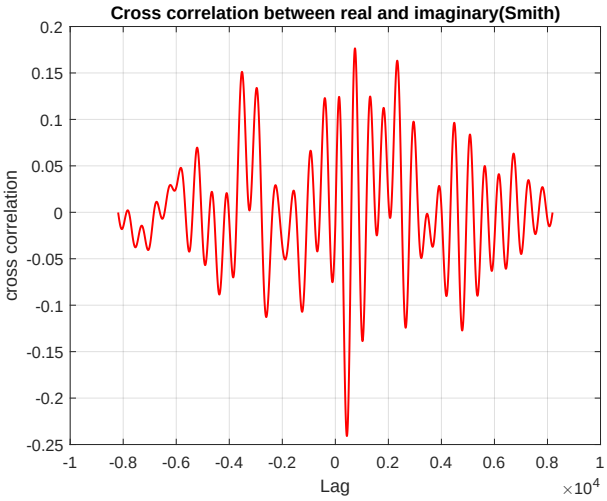
(a) Fading channel magnitude (v = 30m/s)



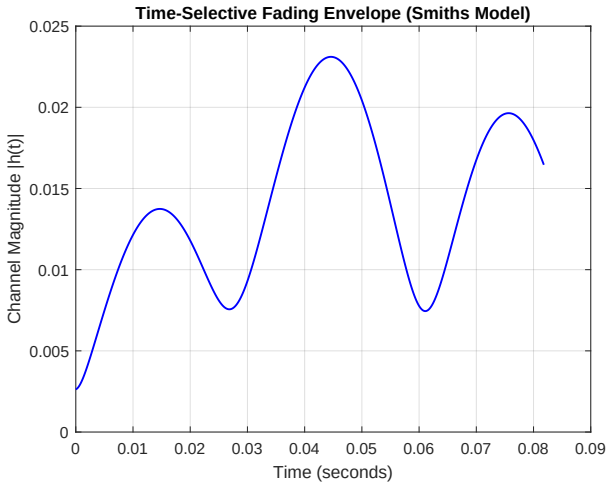
(b) Autocorrelation (Real part)



(c) Autocorrelation (Imaginary part)



(d) Cross Correlation



(e) Fading channel magnitude (v = 3m/s)

Figure 2: Analysis of time-varying fading channel magnitude and correlation properties across different mobilities using Smiths's algorithm.

3 Sum of Sinusoids Model

The Sum of Sinusoids (SoS) model provides an efficient deterministic method to simulate a Rayleigh fading channel by superimposing multiple discrete sine waves to approximate the physical scattering environment. The complex baseband fading envelope $Z(t)$ is defined as:

$$Z(t) = Z_c(t) + jZ_s(t) \tag{1}$$

where the in-phase component $Z_c(t)$ and the quadrature component $Z_s(t)$ are given by:

$$Z_c(t) = \sqrt{\frac{2}{M}} \sum_{n=1}^{M} \cos(\omega_d t \cos \alpha_n + \phi_n) \tag{2}$$

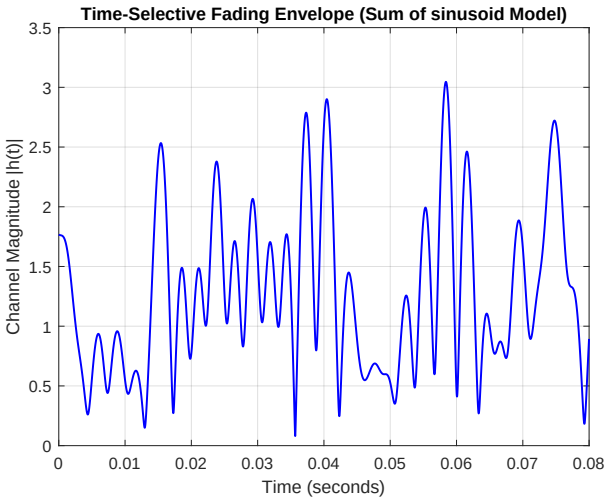
$$Z_s(t) = \sqrt{\frac{2}{M}} \sum_{n=1}^{M} \cos(\omega_d t \sin \alpha_n + \varphi_n) \tag{3}$$

with the discrete angles of arrival, α_n, calculated as:

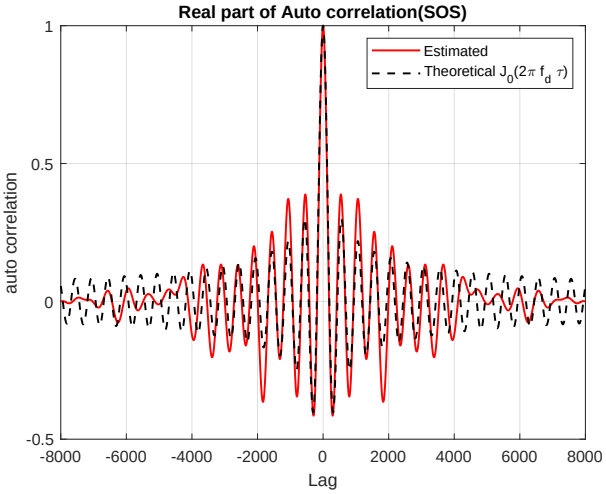
$$\alpha_n = \frac{2\pi n - \pi + \theta}{4M}, \quad n = 1, 2, \ldots, M \tag{4}$$

where ϕ_n, φ_n, and θ are statistically independent random variables, uniformly distributed on $[-\pi, \pi)$ for all n.

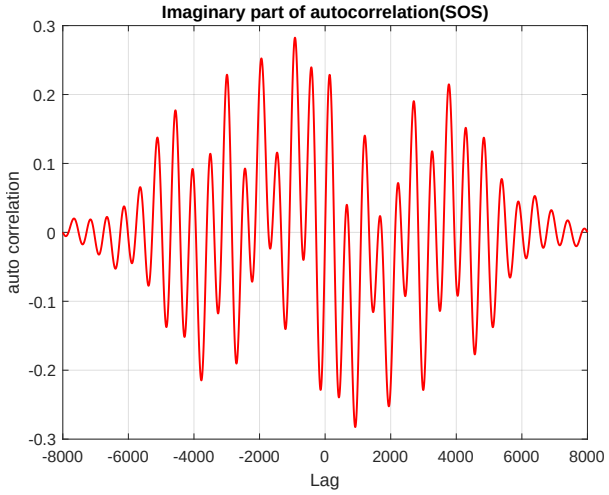
To ensure a highly accurate statistical approximation of the theoretical Rayleigh fading properties and a smooth channel envelope, the number of oscillators was set to $M = 100$ for the purpose of these simulations.



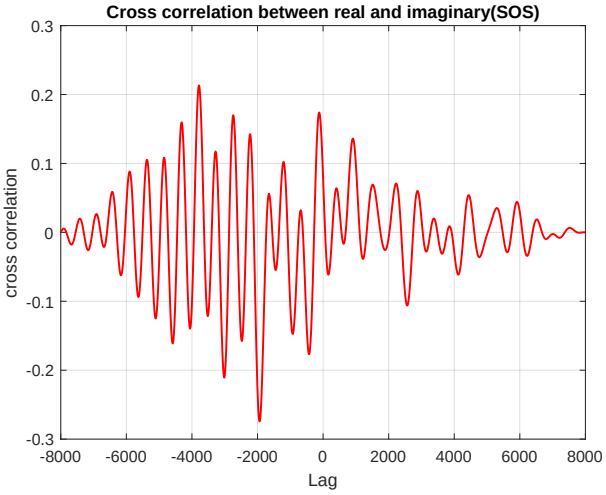
(a) Fading channel magnitude (v = 30m/s)



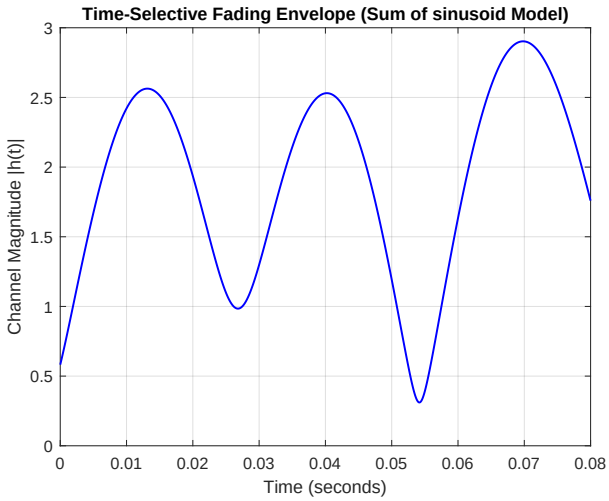
(b) Autocorrelation (Real part)



(c) Autocorrelation (Imaginary part)



(d) Cross Correlation



(e) Fading channel magnitude (v = 3m/s)

Figure 3: Analysis of time-varying fading channel magnitude and correlation properties across different mobilities using Sum of Sinusoids's algorithm.

4 Observations

- We can see the the auto correlation function closely resembles the bessel function of the first kind which is what theoretically we expect.
- The cross correlation oscillates around 0 showing that the expected cross correlation is 0. Which is what we would expect theoretically.
- When we reduce the v from 30m/s to 3m/s we see that the coherence time (how fast it fades) increases this shows a slow fade for smaller doppler frequencies.

5 Frequency Selective fading

This question asks us to simulate a channel with a given power delay profile without any doppler effect. The max delay in samples that we get is 123 so our channel response is a 123 size filter. Now we take a 2048 point FFT of this like the question asked and here are the results for 3 realisations of the channel.
You can see the coherence bandwidtih is around 0.5Mhz.

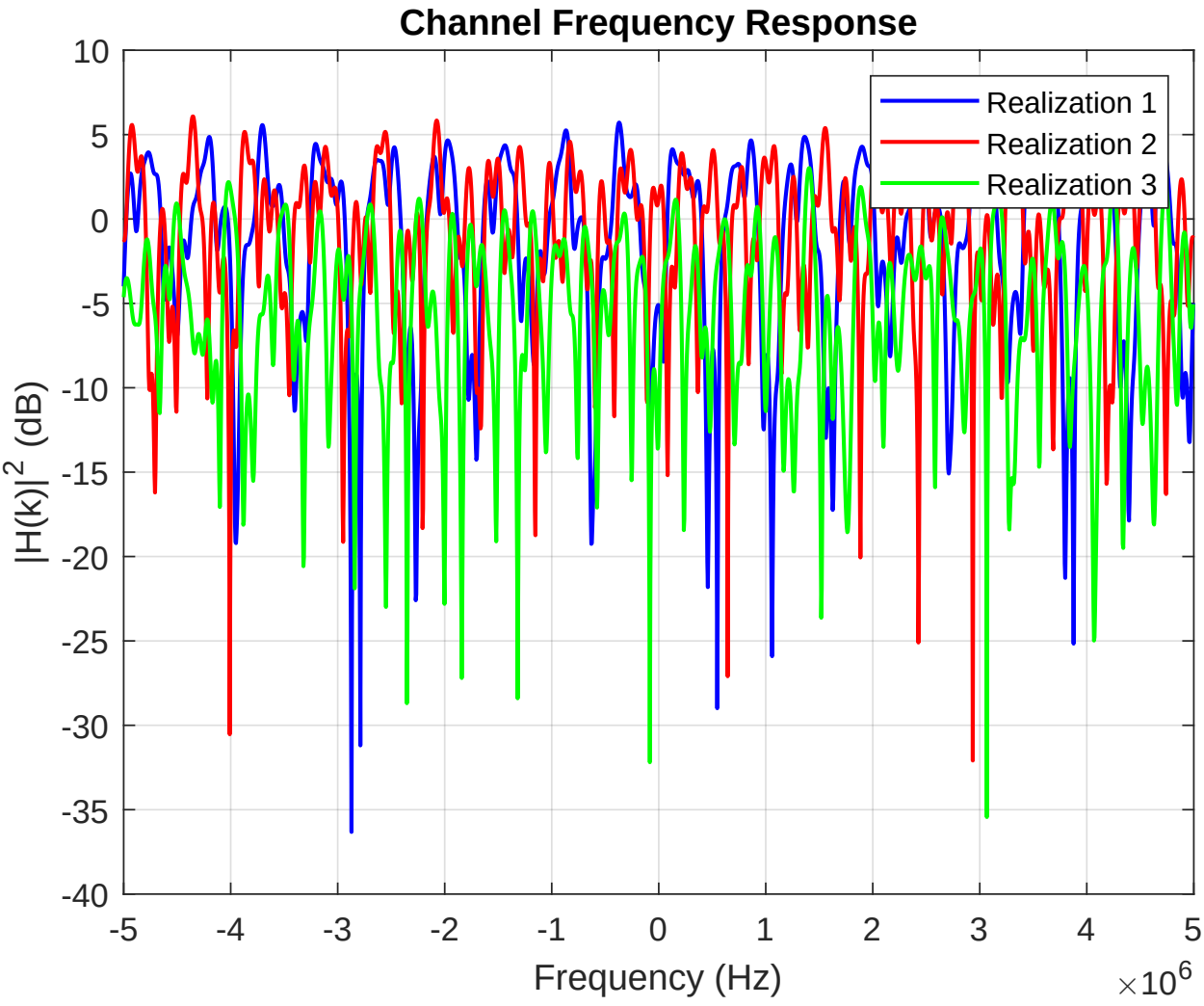


Figure 4: Frequency Selective Channel

6 3D Time varying Frequency selective fading

For each delay I simulated a sum of sinusoidal model with M=100 and fd = 90Hz. Finally got a 2d array for the channel response of size 123 * 7. The 3D plot for the frequency response of the channel is given below.

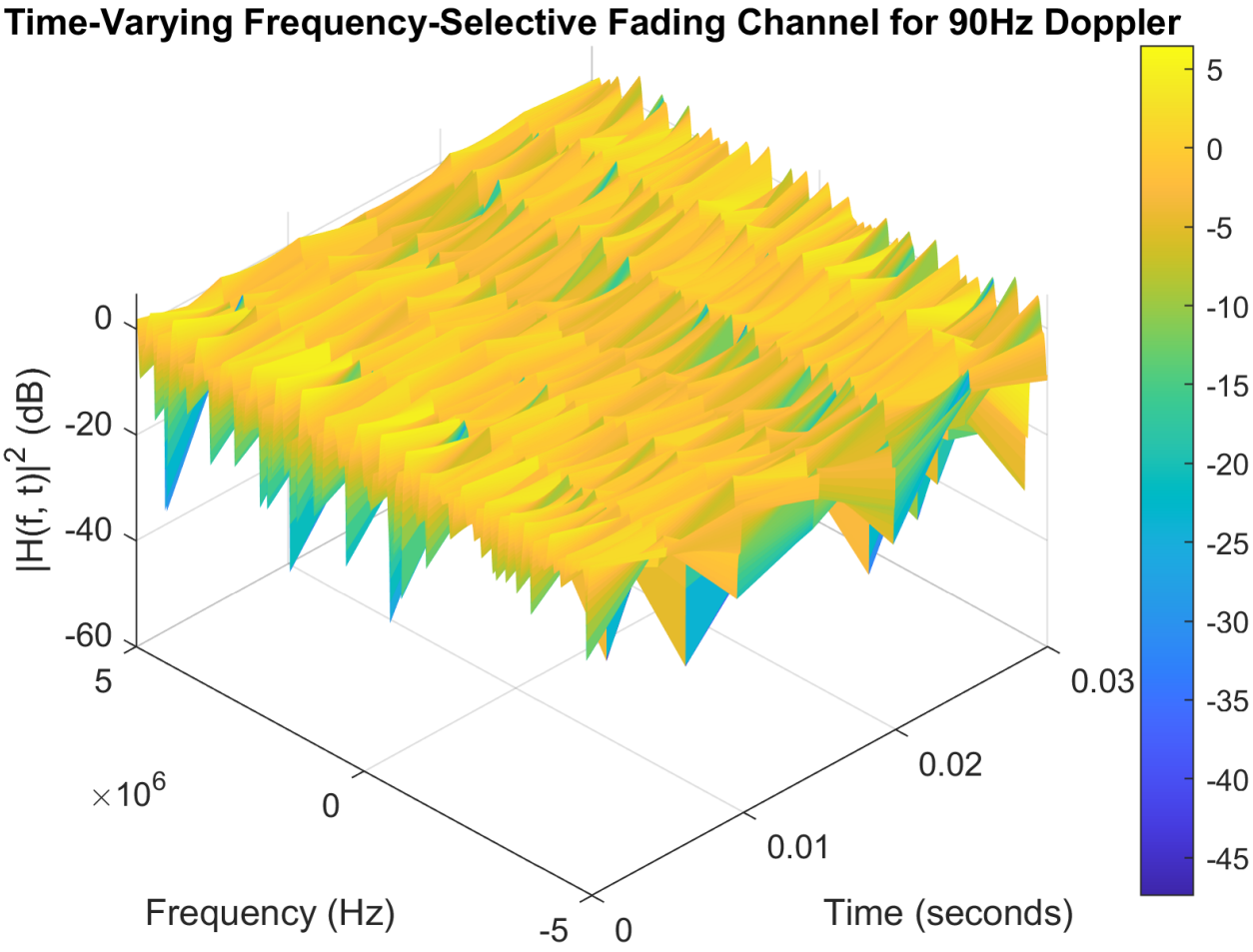


Figure 5: 3D Plot of time varying frequency selective channel